



LLMs in Practice

Lecture 12 part I

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About me



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Outline

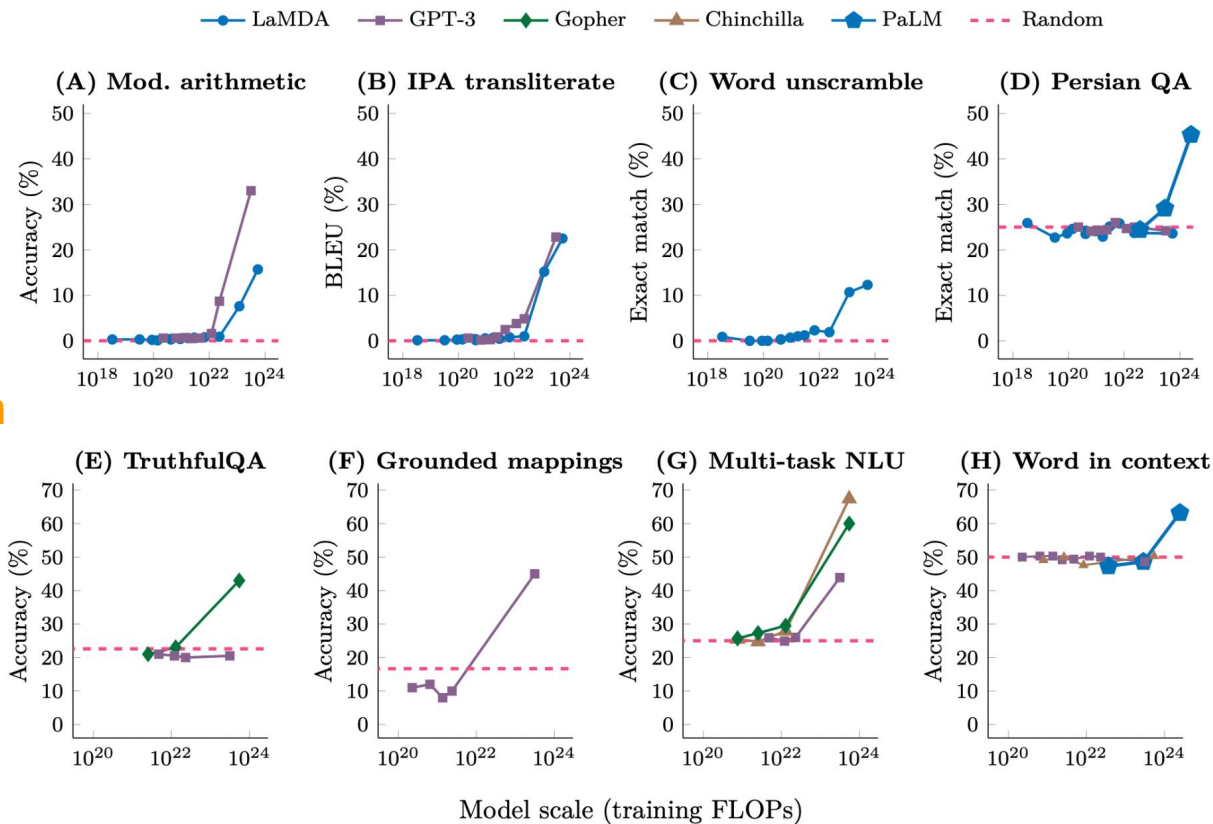
1. Prompting
2. RAG
3. Parameter-efficient fine-tuning
4. Supervised fine-tuning
5. Full Fine-tuning
6. Data Quality & Evaluation
7. Method Decision Framework

least effort -> most effort

Effort: computational costs (time + money + hardware requirements), engineering work, data requirements

Emergent Capabilities

Figure from [Wei et al. \[2022\]](#)



Emergent capabilities in language models are those abilities that are not observed in smaller models but manifest only when the models reach a certain size

Prompting: why it matters?

We demonstrate that GPT-3 can perform tasks with **no gradient updates or fine-tuning**, sometimes reaching performance competitive with fine-tuned models.

[Brown et al. \[2020\]](#)

- Modern LLMs already encode many latent capabilities
- Prompting accesses these capabilities without training
- Cheap, fast, no infrastructure

In-context learning: the model learns a task based on inputs (or prompts) given at inference time, **without updating its parameters**.

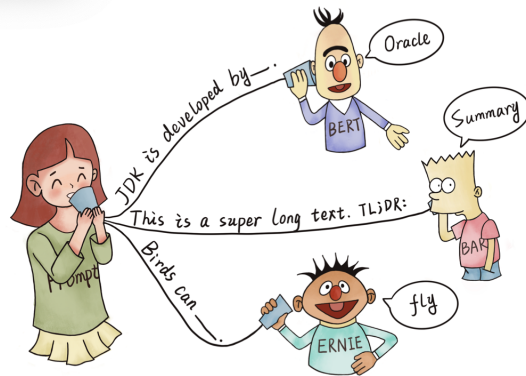


Figure from [Liu et al. \[2021\]](#)

Elements of the prompt

external information or additional context that can steer the model to better responses

Context

You are an expert sentiment analyser

Classify given text into positive, negative and neutral

Text: I enjoy NLP1

Sentiment:

a specific task or instruction you want the model to perform

Instruction

the input that we are interested to find a response for

Input Data

the type of format for the output

Output Indicator

Zero-shot Prompting

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

(Output) 8 **X**

Figure from [Kojima et al. \[2023\]](#)

Simplest way to prompt, instructions only, no examples

Works well when the task is clearly described and similar to patterns seen during pretraining and the output space is constrained

Guides prompt design before spending time on data collection or fine-tuning.

Few-shot Prompting

Figure from [Kojima et al. \[2023\]](#)

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

→ example

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The answer is 8. **X**

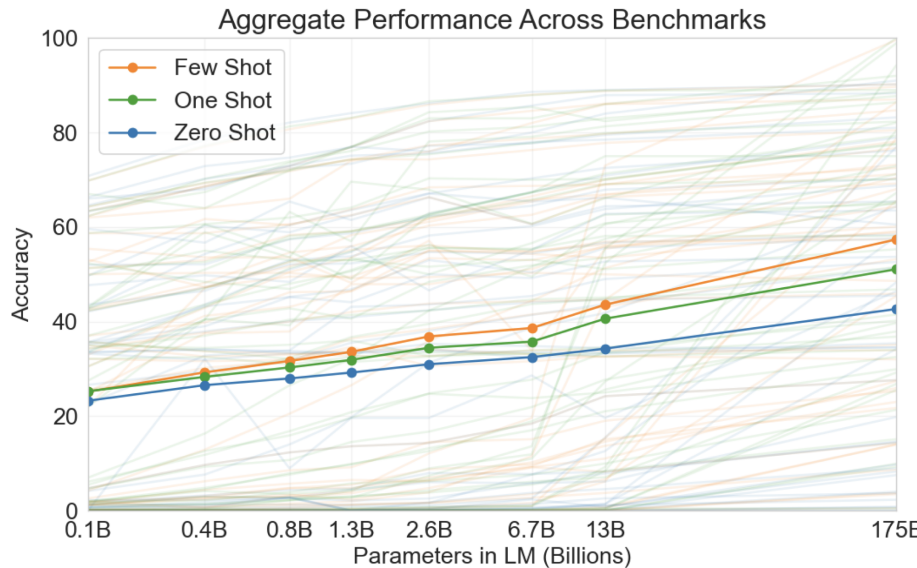
Helps when task is underspecified, the output needs a specific structure, labels are non-obvious

Figure from [Brown et al \[2020\]](#)

Introduced by [Brown et al.\[2020\]](#) from OpenAI

One-shot -> one example included in the prompt

Two-shot -> two examples etc...



Chain-of-Thought Prompting

Introduced by [Wei et al. \[2023\]](#) from the Google Research team

The model is explicitly asked to explain its reasoning steps (chain of thought) before the answer

“Let’s think step by step.”

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) *The juggler can juggle 16 balls. Half of the balls are golf balls. So there are $16 / 2 = 8$ golf balls. Half of the golf balls are blue. So there are $8 / 2 = 4$ blue golf balls. The answer is 4. ✓*

In this example they combine one-shot with CoT

Figure from [Kojima et al. \[2023\]](#)

Alternative Prompting Techniques I

Role-play Prompting: Assign a role to the LLM (“You are a clinical psychologist. Answer as you would in a report.”) [Kong et al. \[2024\]](#)

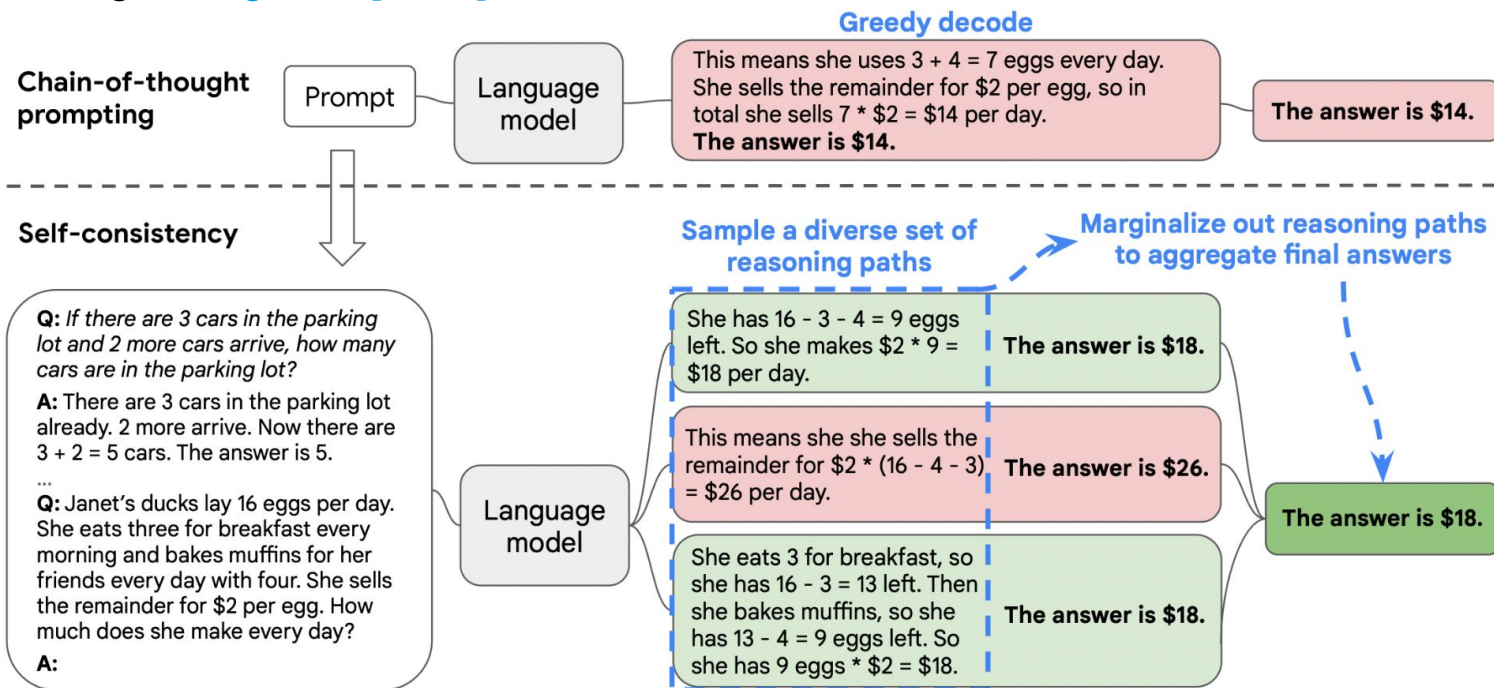
Meta Prompting: Prompting technique that emphasizes the structural and syntactical aspects of problems by prioritizing the overall format and pattern over specific content details. [Zhang et al. \[2024\]](#)

Integrate step-by-step reasoning to solve mathematical problems under the following structure:

```
{
  "Problem": "[question to be answered]",
  "Solution": {
    "Step 1": "Begin the response with “Let’s think step by step.””,
    "Step 2": "Follow with the reasoning steps, ensuring the solution process is broken down clearly and logically.",
    "Step 3": "End the solution with the final answer encapsulated in a LaTeX-formatted box,  $\boxed{\dots}$ , for clarity and emphasis."
  },
  "Final Answer": "[final answer to the problem]"
}
```

Alternative Prompting Techniques II

Self-consistency COT: Generate multiple diverse chain-of-thoughts via sampling and take the majority answer, replacing the single, brittle reasoning path produced by greedy decoding. [Wang et al \[2022\]](#)



Limitations of Prompting

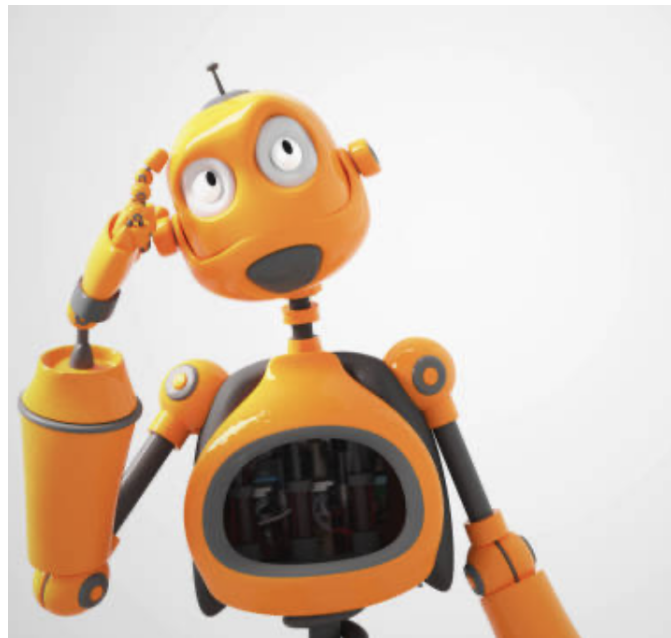
Prompting changes the *input*, not the *model*.

When the model lacks knowledge, stability, or skills, prompting hits a ceiling.

Cannot add new knowledge

Unstable behaviour

No way to enforce domain style





Problem Statement



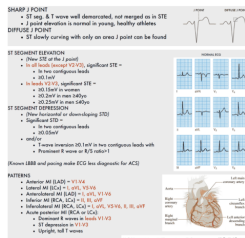
Medtronic

SIEMENS
Healthineers

Medical device manufacturing and service company



ECG machine malfunction



Thousands of pages of internal documents (manuals)

ChatGPT

Hmm...something seems to have gone wrong.

< 2 / 2 >   

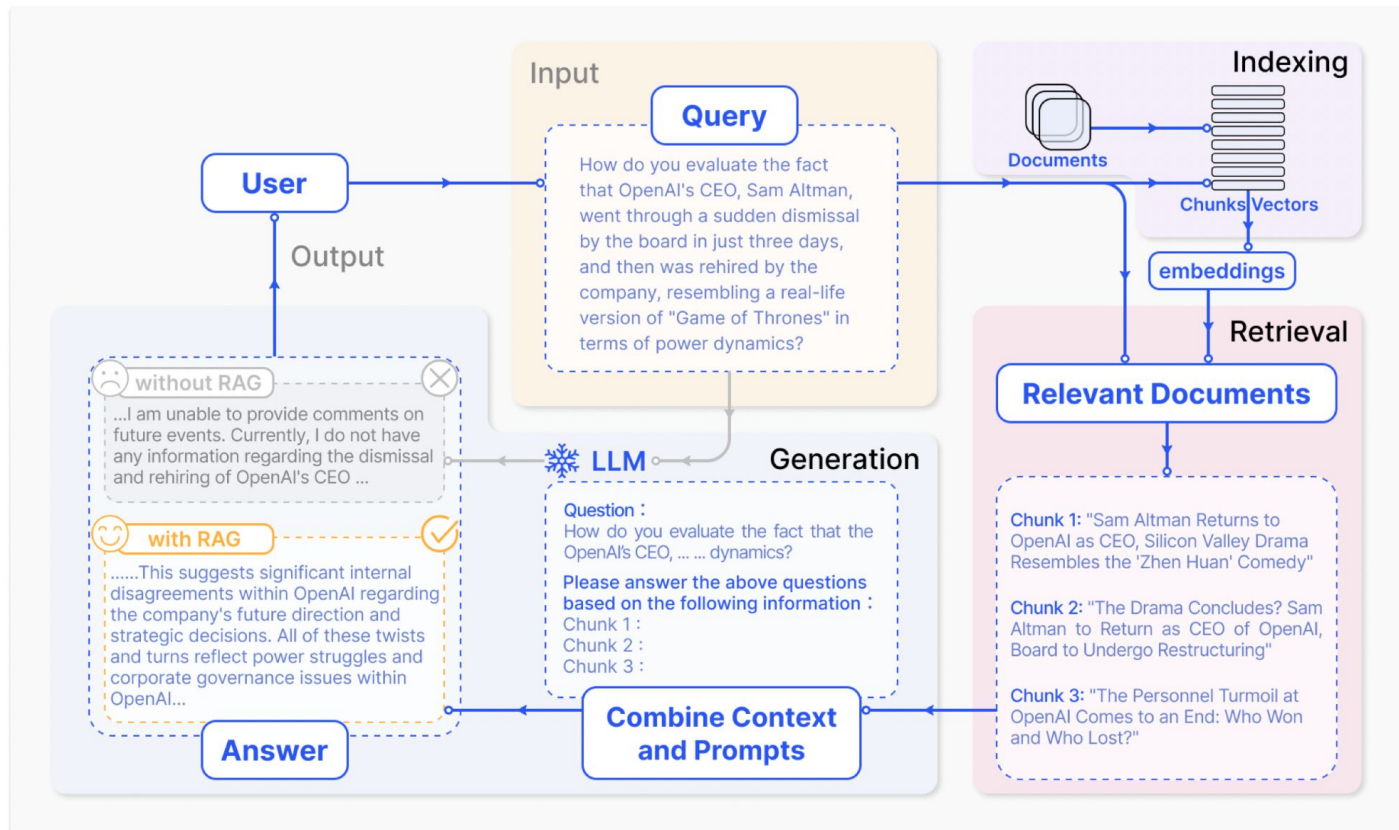
There was an error generating a response

Regenerate

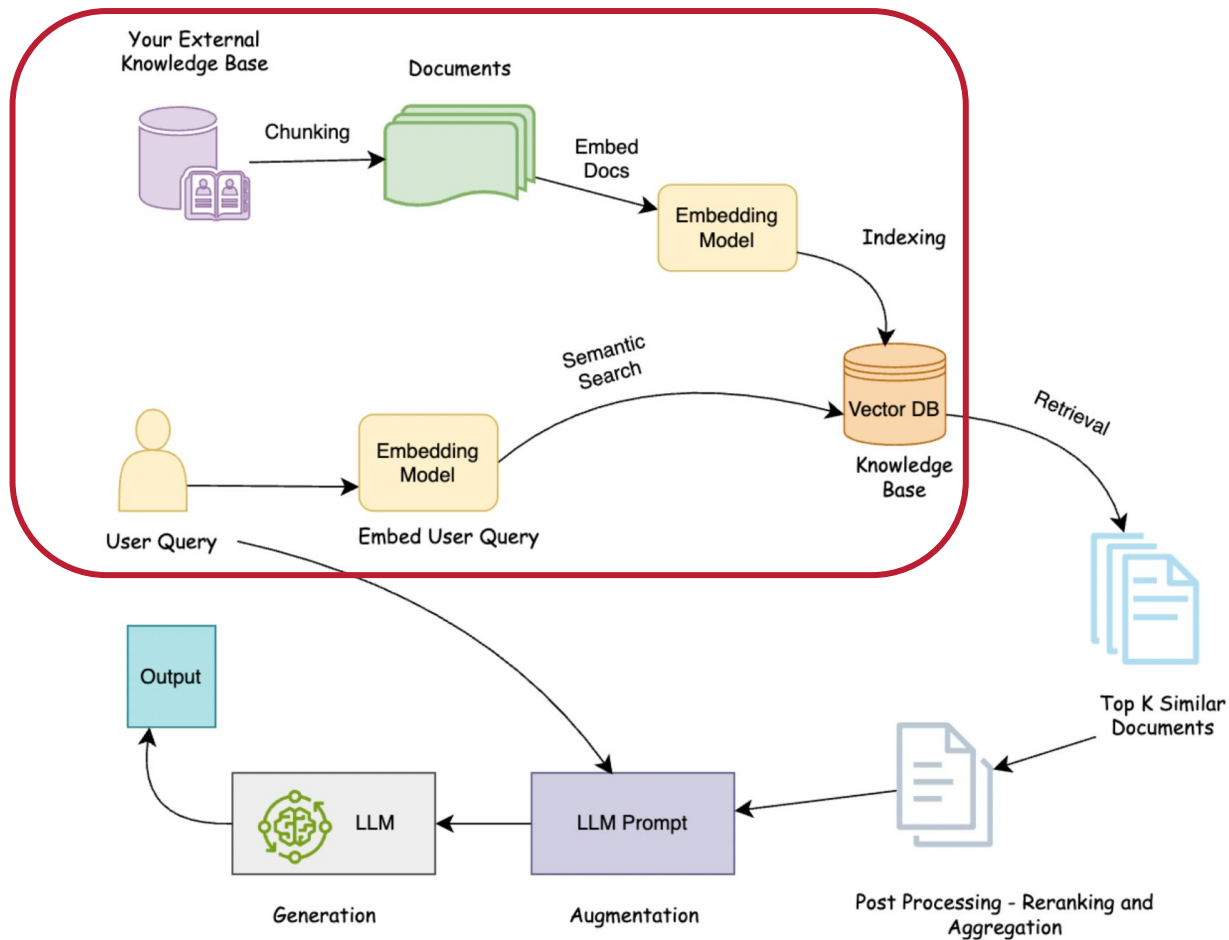
ChatGPT cannot draw knowledge from proprietary documents

Possible Solution - RAG

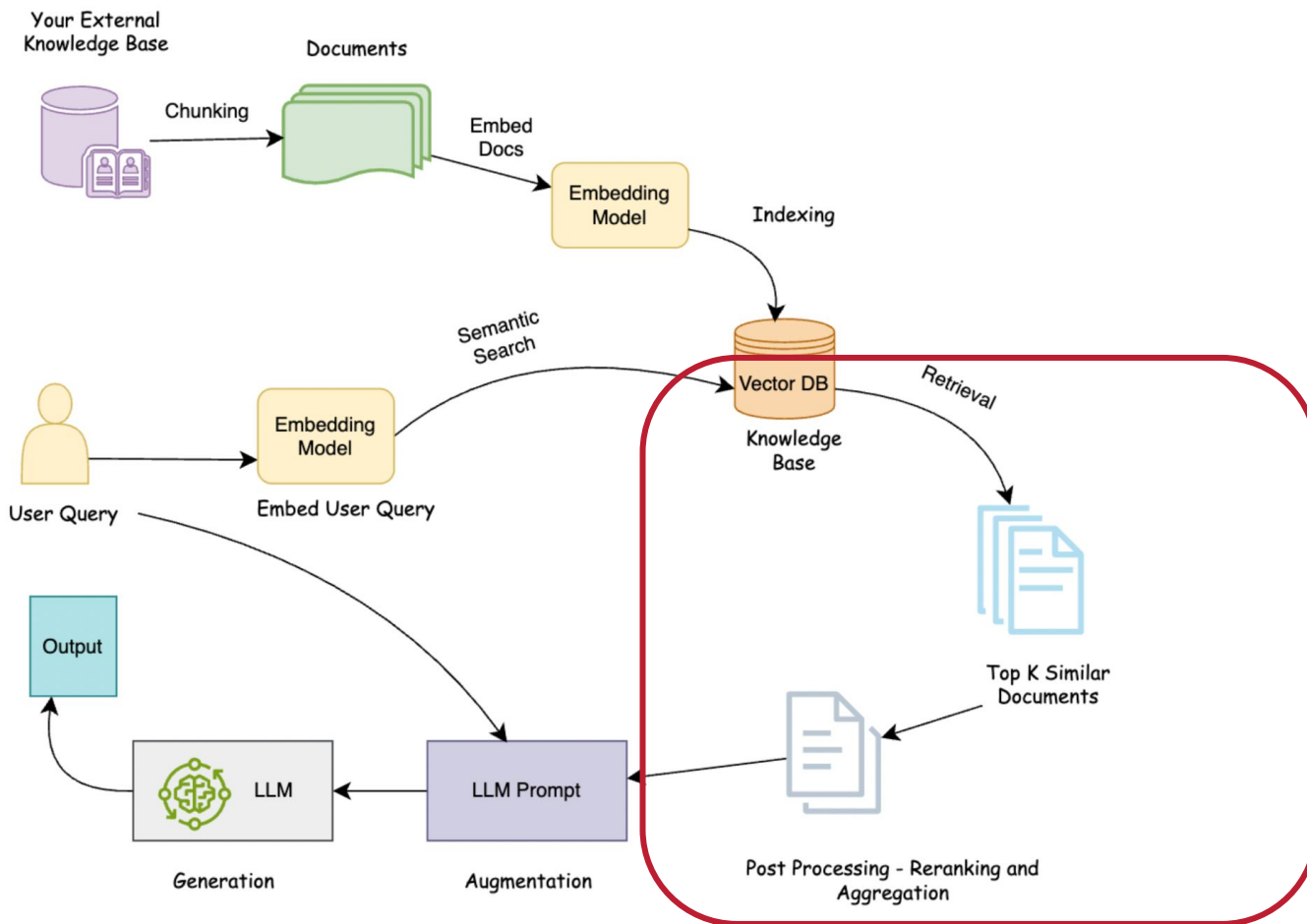
Retrieval Augmented Generation [Lewis et al \[2020\]](#)



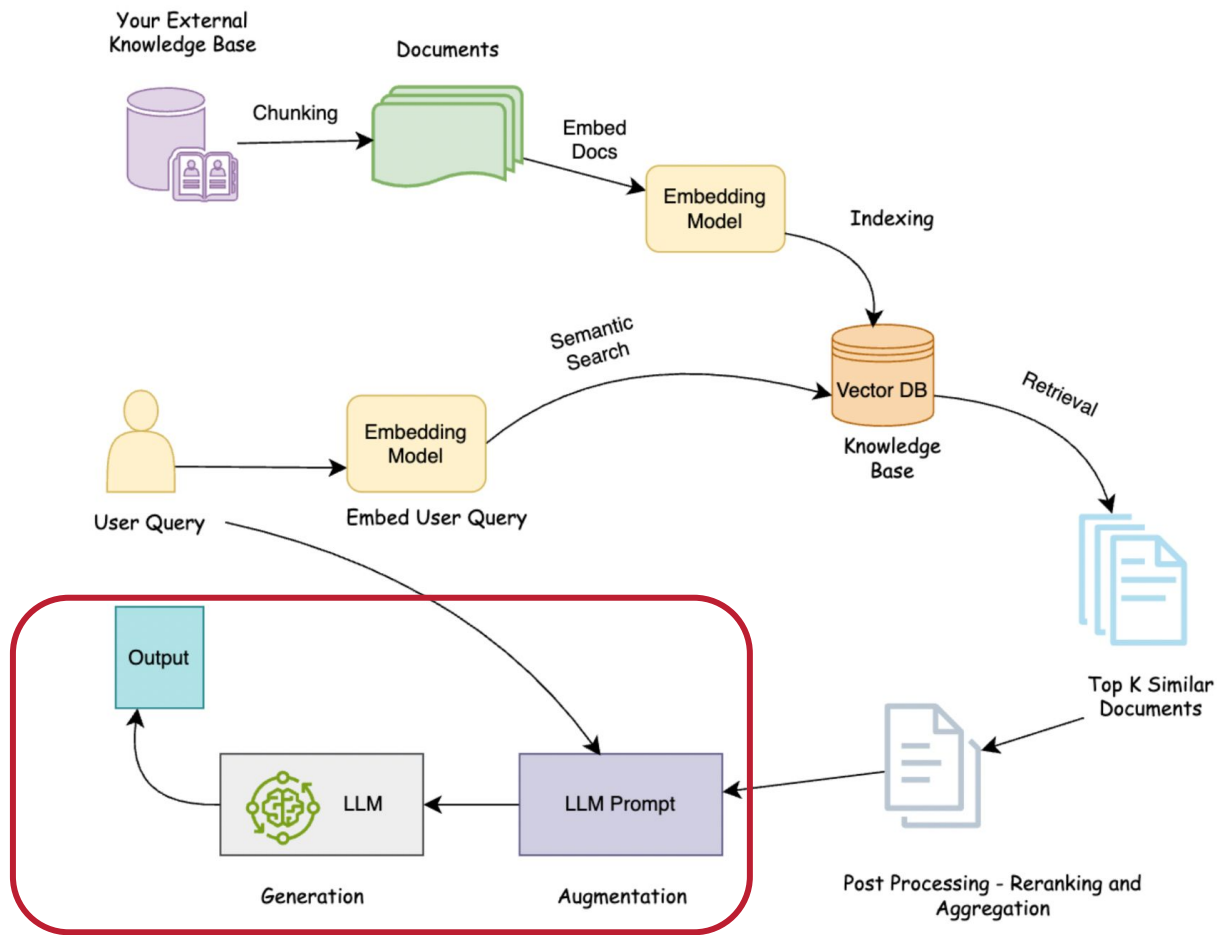
Indexing



Retrieval

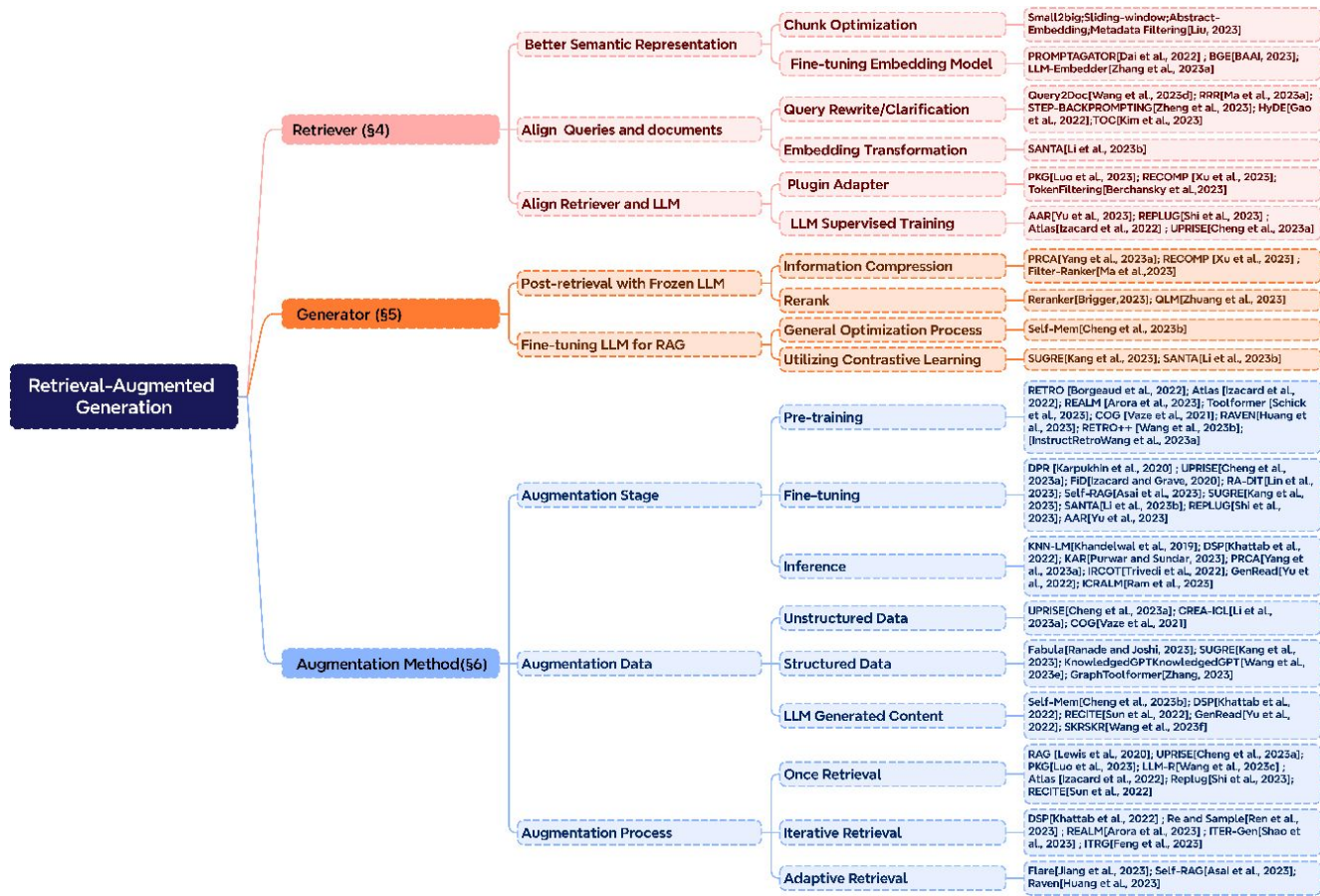


Generation



Different Architectures

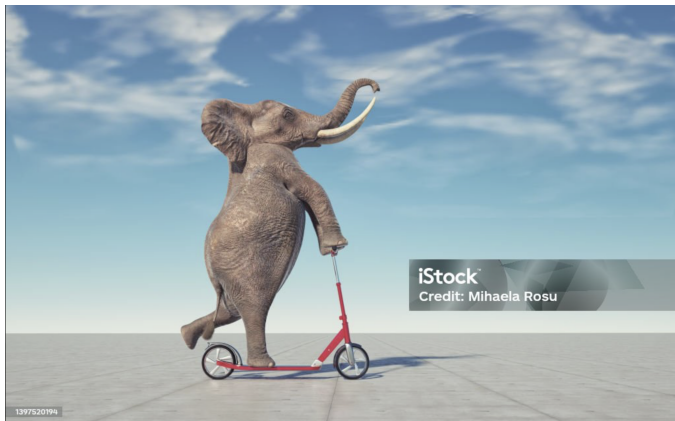
Retrieval Augmented Generation for Large Language Models: A survey ([Gao et al. \[2024\]](#))



Problem Persists

Knowledge is not the only limitation

We have the right knowledge, but not the right behaviour



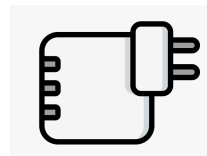
```
{
  glossary: {
    title: "example glossary",
    GlossDiv: {
      title: "S",
      GlossList: {
        GlossEntry: {
          ID: "SGML",
          SortAs: "SGML",
          GlossTerm: "Standard Generalized Markup Language",
          Acronym: "SGML",
          Abbrev: "ISO 8879:1986",
          GlossSee: "markup"
        }
      }
    }
  }
}
```

Parameter Efficient Fine-tuning

Adapting an LLM's behaviour without retraining the model

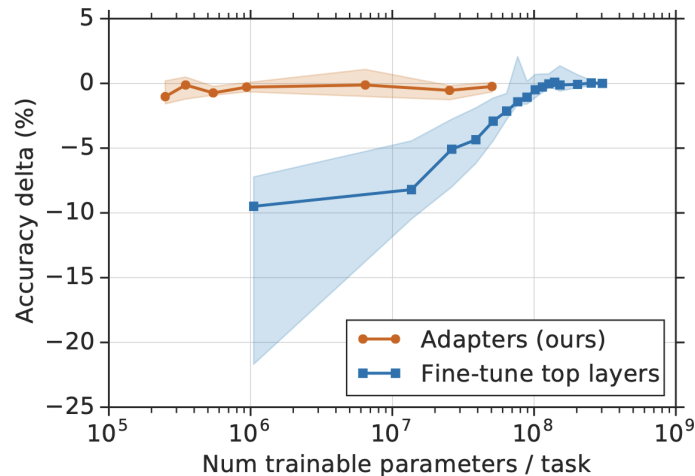
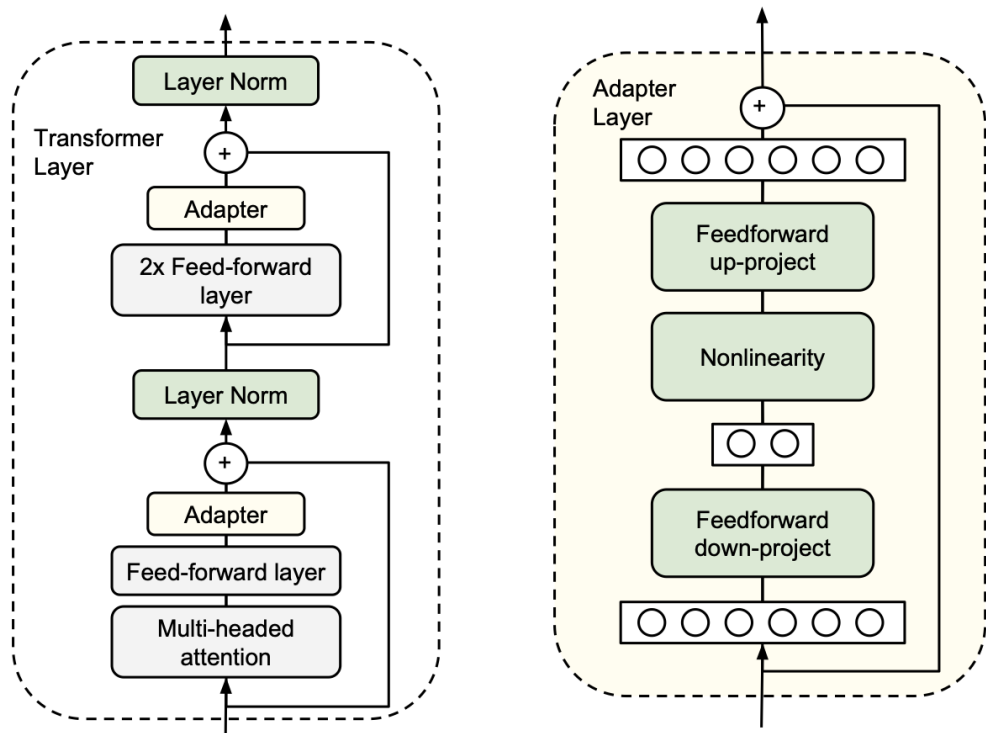
Learn small components that help 'adapt' the model

- Freeze all original LLM weights
 - Add tiny trainable modules (<1% of parameters) or rethink the way weight matrices are used
 - Learn task-specific behaviour from small datasets
-
- Fits on consumer-grade GPUs (even single-GPU setups)
 - Requires much less memory (small model state)
 - Modules are lightweight add-ons that can be swapped or combined
-
- Reduces risk of catastrophic forgetting
 - Lower chance of overfitting on small domain datasets



Adapters

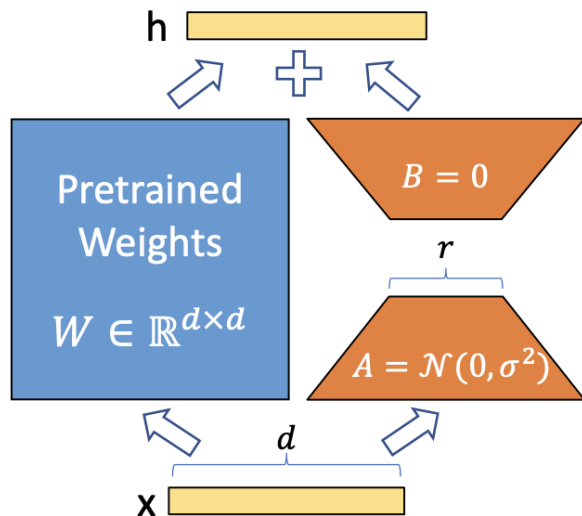
Parameter-Efficient Transfer Learning for NLP [Houlsby et al. \[2019\]](#)



See also Adapter Fusion ([Pfeiffer et al. \[2020\]](#)): independently train various adapters then learn to ‘fuse’ them (combine them via a trained attention mechanism).

LoRA

LoRA: Low-Rank Adaptation of LLMs ([Hu et al. \[2021\]](#))



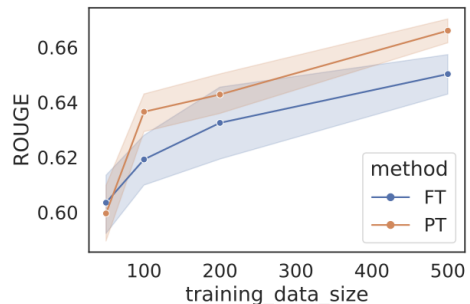
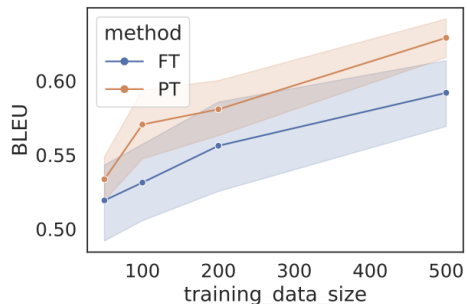
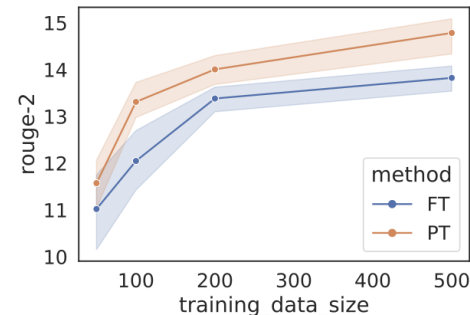
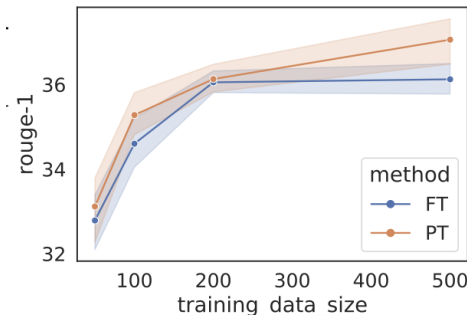
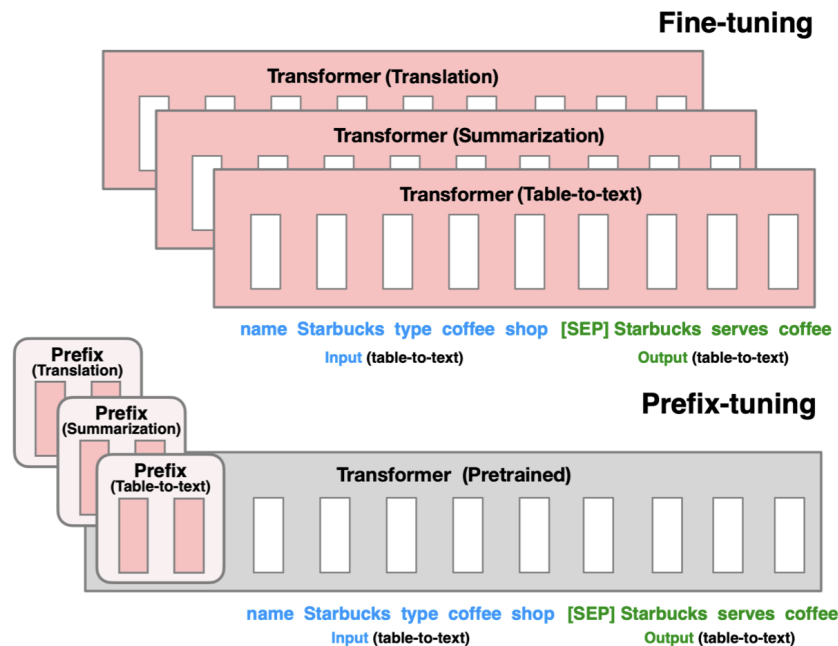
$$W_0 + \Delta W = W_0 + BA$$

Model&Method	# Trainable Parameters	WikiSQL	MNLI-m	SAMSum
		Acc. (%)	Acc. (%)	R1/R2/RL
GPT-3 (FT)	175,255.8M	73.8	89.5	52.0/28.0/44.5
GPT-3 (BitFit)	14.2M	71.3	91.0	51.3/27.4/43.5
GPT-3 (PreEmbed)	3.2M	63.1	88.6	48.3/24.2/40.5
GPT-3 (PreLayer)	20.2M	70.1	89.5	50.8/27.3/43.5
GPT-3 (Adapter ^H)	7.1M	71.9	89.8	53.0/28.9/44.8
GPT-3 (Adapter ^H)	40.1M	73.2	91.5	53.2/29.0/45.1
GPT-3 (LoRA)	4.7M	73.4	91.7	53.8/29.8/45.9
GPT-3 (LoRA)	37.7M	74.0	91.6	53.4/29.2/45.1

See also QLoRA: Efficient Finetuning of Quantised LLMs ([Dettmers et al. \[2023\]](#))

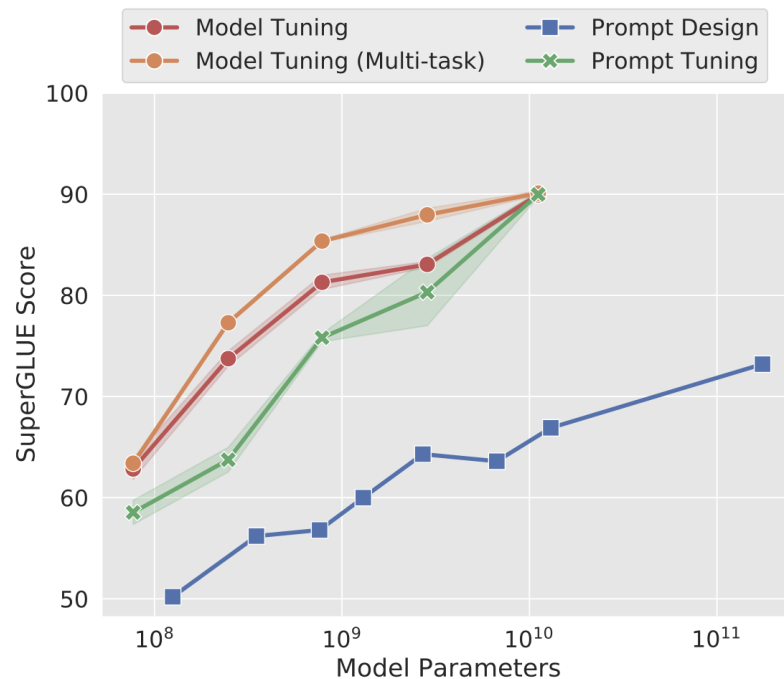
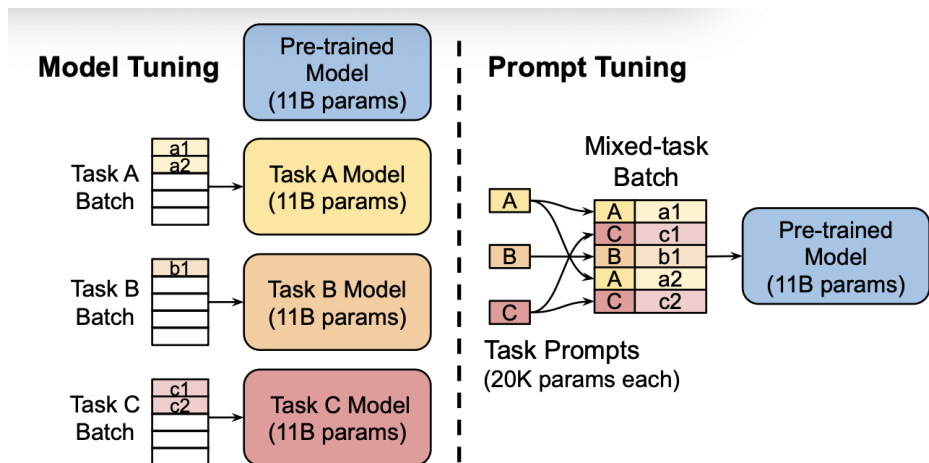
Prefix Tuning

Prefix Tuning: Optimizing Continuous Prompts for Generation [Li et al. \[2021\]](#)



Prompt Tuning

The Power of Scale for Parameter-Efficient Prompt Tuning ([Lester et al. \[2021\]](#))



Prompt tuning keeps the model frozen and learns only a small set of continuous “prompt” embeddings prepended to the input, steering the model toward a new task with minimal extra parameters.

Instruction Tuning

Teach the model to follow instructions using supervised examples

Finetuned Language Models are Zero-Shot Learners ([Wei et al. \[2021\]](#))

Finetune on many tasks (“instruction-tuning”)

Input (Commonsense Reasoning)
 Here is a goal: Get a cool sleep on summer days.
 How would you accomplish this goal?
 OPTIONS:
☐ -Keep stack of pillow cases in fridge.
☐ -Keep stack of pillow cases in oven.
Target
☒ keep stack of pillow cases in fridge

Input (Translation)
 Translate this sentence to Spanish:
 The new office building was built in less than three months.
Target
 El nuevo edificio de oficinas se construyó en tres meses.

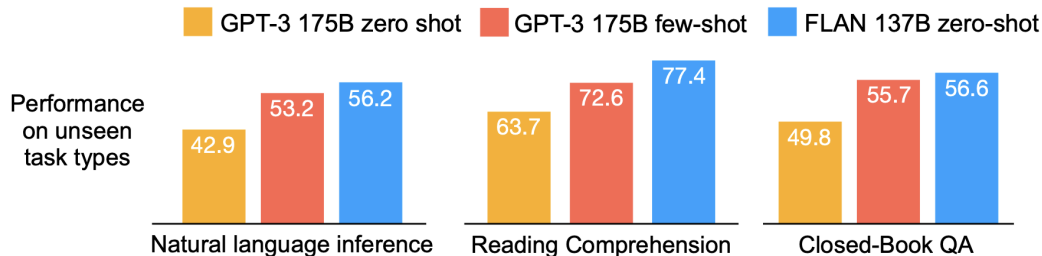
Sentiment analysis tasks
 Coreference resolution tasks
 ...

Inference on unseen task type

Input (Natural Language Inference)
 Premise: At my age you will probably have learnt one lesson.
 Hypothesis: It's not certain how many lessons you'll learn by your thirties.
 Does the premise entail the hypothesis?
 OPTIONS:
☒ -yes ☐ -it is not possible to tell ☐ -no
FLAN Response
☒ It is not possible to tell

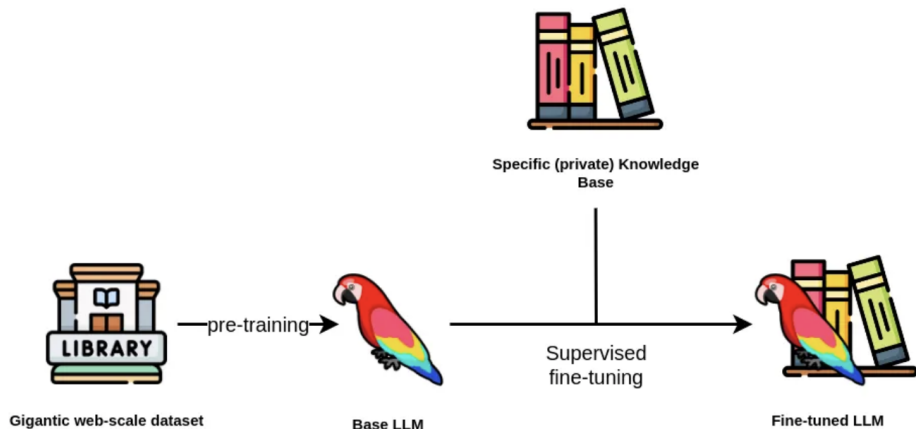
Trained on (instruction,response) pairs

Helps the model internalise general instruction-following behaviour



Supervised Fine-tuning

Teach the model a specific task from labelled examples

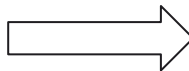


[image source](#)

- Take a pretrained LLM
- Train it further on a smaller dataset of (input -> target output pairs)
- Optimise a supervised loss (usually cross-entropy on next tokens)
- The model learns to imitate the desired behaviour of the task

SFT Example

Patient reports a 3-day history of worsening shortness of breath and dry cough. No fever, but mild wheezing noted on exam. Past history significant for asthma. Started on inhaled bronchodilator with good initial response. Plan to reassess in 48 hours



```
{
  "chief_complaint": "Worsening shortness of breath and dry cough for 3 days",
  "history_of_present_illness": {
    "duration": "3 days",
    "symptoms": [
      "shortness of breath (worsening)",
      "dry cough",
      "no fever",
      "mild wheezing"
    ],
    "relevant_history": "Asthma"
  },
  "physical_exam": {
    "respiratory": "Mild wheezing noted; no signs of fever"
  },
  "diagnosis": {
    "description": "Asthma exacerbation",
    "icd10_code": "J45.909"
  },
  "treatment": {
    "medications": [
      {
        "name": "Inhaled bronchodilator",
        "response": "Good initial response"
      }
    ]
  },
  "plan": {
    "follow_up": "Reassess in 48 hours"
  }
}
```

a few thousand examples of clinician note -> correct structured example

- outputs invalid JSON
- adds or removes fields
- invents diagnoses that were never mentioned
- hallucinates icd10 codes
- writes disclaimers ("I am not a doctor")

prompt

SYSTEM: You convert free-text clinical notes into structured EHR JSON.
NOTE: Patient reports a 3-day history of worsening shortness...
OUTPUT:

model predicts tokens



Compute loss between
predicted tokens and
target JSON



backpropagate
update weights

Full Fine-tuning

Update **100%** of the model parameters

you can teach the model new reasoning abilities, new knowledge, new domains
requires GPU Clusters (very expensive)
risk of catastrophic forgetting

Who does that?



OpenAI

ANTHROPIC



Meta

Large research labs , very large companies (Bloomberg, DataBricks)

Just do PEFT (95-100 percent of the performance on <1% of the cost and risk)

Fine-Tuning Requirements

- Let's fine-tune a $\Psi=7.5\text{B}$ Transformer using Adam with mixed precision

We need 16Ψ GB (=120 GB) of memory for the “model states” alone:

- **Model:** 2Ψ bytes (for fp16 parameters)
- **Gradients:** 2Ψ bytes (for fp16 parameters)
- **Optimiser:** 12Ψ (for fp32 parameters, momentum and variances)

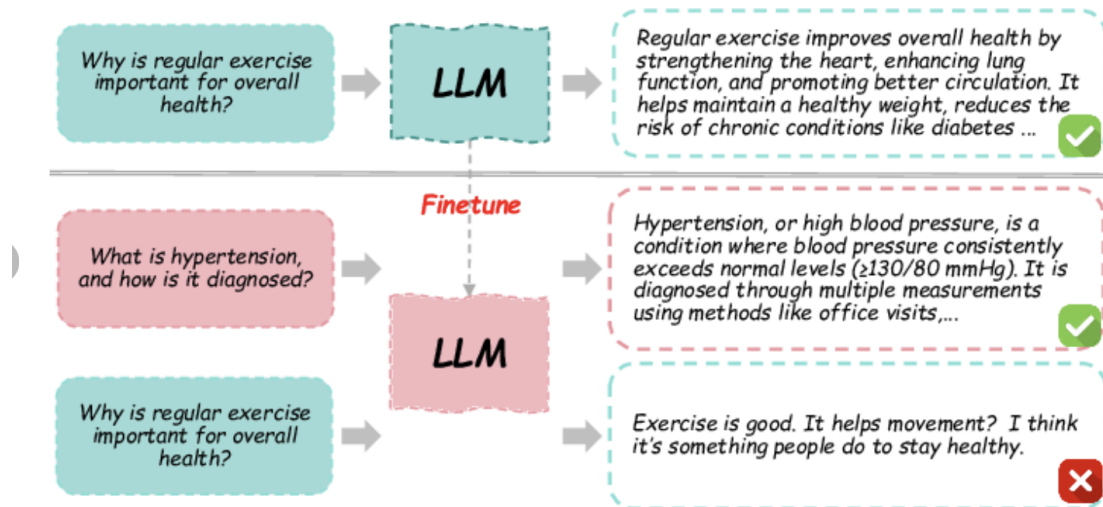
- **Activations**

For a GPT2 architecture (NumLayers=48 Layers, HiddenDim=1600 units) with SeqLen=1K and BatchSize=32, we need about 60 GB of memory

Solution -> Multi-GPU training

Catastrophic Forgetting

Neural networks could overwrite older knowledge when trained on new data



[image source](#)

- can affect reasoning, language understanding etc
- reason to continuously evaluate, even on previous benchmarks
- PEFT does not suffer from this limitation

Data Quality & Evaluation

Most LLM failures come from bad data and weak validation, not from model choice

Data Quality Problems

- inconsistent labels
- missing edge cases
- poor formatting
- non-representative dataset

Why evaluation is hard

- correctness is not always binary
- annotations can be expensive
- sensitivity to formatting or phrasing
- difficult to define ‘ground truth’

Evaluation Good Practices

- holdout validation split
- gold-standard test set curated by human experts
- LLM judge (?) with human-in-the-loop
- task-specific metrics

Knowledge from the streets

- 80% of time dealing with data
- 15% of time writing evaluations
- 5% of the time coding, configuring

See also A survey on Evaluation of Large Language Models [Chang et al. \[2023\]](#)

Decision Framework

How to Decide: Prompting, RAG, PEFT, or Full FT?

Prompting

- *don't use a bulldozer to kill a fly*
- task is light
- you need behaviour style changes
- no strict constraints
- no proprietary data needed

RAG

- the model lacks the needed knowledge
- you cannot finetune (frequent data updates)
- you want explainable, source-grounded answers

PEFT

- you need consistent behaviour, enforcement of output structure
- dataset size is small - medium (<100k samples)
- you want low cost, minimal risk of catastrophic forgetting

FFT

- you are operating at foundation-model scale
- you need new deep-level capabilities
- you have huge datasets (millions to billions of tokens)



Acknowledgements

Part of the slides was inspired by Wilker Aziz

Practical Tips & Tricks

larger model not always better $\implies$ try Llama-3.1-8B $\implies$ consider quantization

be wary of [licencing](#) (Apache 2.0, MIT)

keep the context window limitations in mind

keep the [billing](#) in mind \$\$

design according to [application](#) (inference speed, answer truthfulness)